Lista Encadeada

Estruturas de Dados

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Uso de arrays em estruturas lineares

Acesso direto (eficiência)

Alocação em tempo de compilação (desperdício de memória)

Listas ligadas (linked lists)

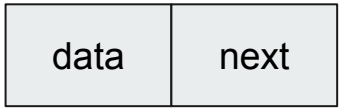
TAD fundamental

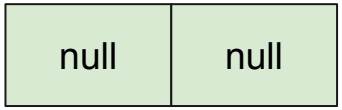
Tamanho arbitrário e dinâmico

Acesso sequencial

Permite elementos duplicados

Listas ligadas (Design)

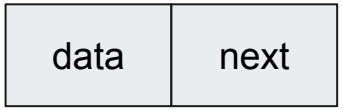




head

head

Listas ligadas (Design)



head

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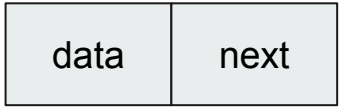
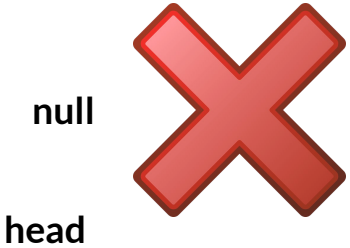
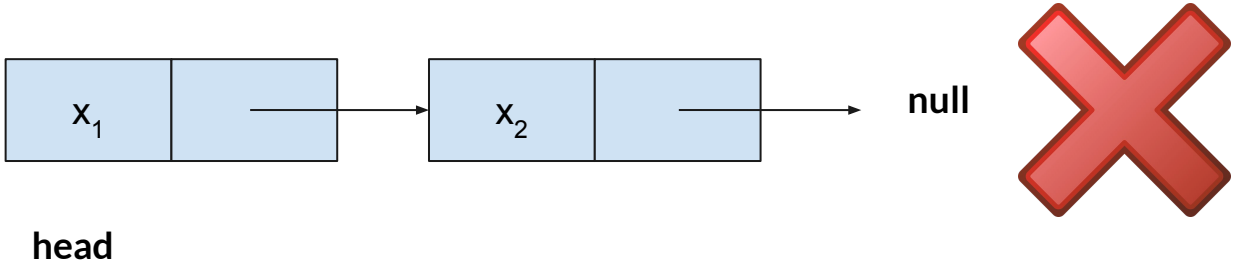
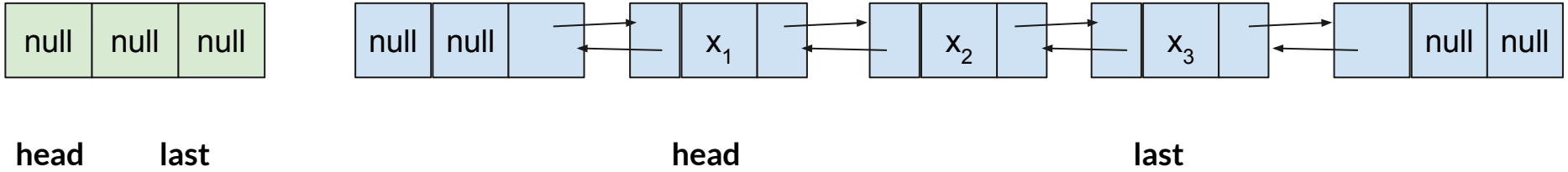
Listas ligadas (Design)



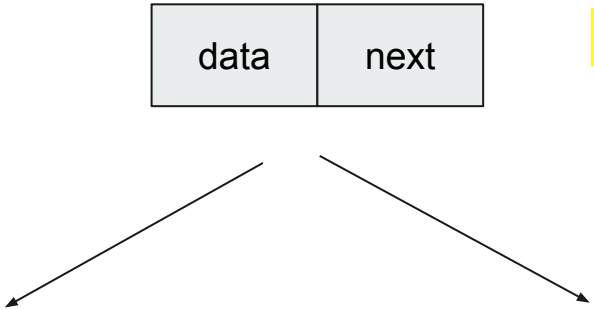




Listas duplamente ligadas (Design)



Listas ligadas (Design)



Estrutura recursiva!

Abordagem iterativa

Abordagem recursiva

Listas ligadas (paradigma funcional)

Listas ligadas (operações)

- Inserir
- Remover
- Pesquisar
- Tamanho
- Vazio

Listas ligadas (simulação)

Lista Encadeada
Abordagem Iterativa

Listas ligadas - implementação

```java
public interface LinkedList<T> {

    public void insert(T element);
    public void remove(T element);
    public T search(T element);
    public int size();
    public boolean isEmpty();
    public T[] toArray();
}
```

```java
public interface DoubleLinkedList<T>
    extends LinkedList<T> {

    public void insertFirst(T element);
    public void removeFirst();
    public void removeLast();

}
```

Listas ligadas - implementação

data	next

```
public class SingleLinkedListNode<T> {
  protected T data;
  protected SingleLinkedListNode<T> next;

  public SingleLinkedListNode() { }

  public SingleLinkedListNode(T data,
      SingleLinkedListNode<T> next) {
     this.data = data;
     this.next = next;
  }
  … gets e sets
}
```

previous	data	next

```
public class DoubleLinkedListNode<T>
      extends SingleLinkedListNode<T> {

  protected DoubleLinkedListNode<T> previous;
  public DoubleLinkedListNode() { }
  public DoubleLinkedListNode(T data,
           DoubleLinkedListNode<T> next,
           DoubleLinkedListNode<T> previous) {
    super(data, next);
    this.previous = previous;
  }

  … gets e sets
}
```

Listas ligadas - implementação

```
public class SingleLinkedListNode<T> {
  protected T data;
  protected SingleLinkedListNode<T> next;

  public SingleLinkedListNode() { }  NIL

  public SingleLinkedListNode(T data,
      SingleLinkedListNode<T> next) {
     this.data = data;
     this.next = next;
  }
  … gets e sets
}
```

```
public class DoubleLinkedListNode <T>
      extends SingleLinkedListNode<T> {

  protected DoubleLinkedListNode<T> previous;
  public DoubleLinkedListNode() { } NIL
  public DoubleLinkedListNode(T data,
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    super(data, next);
    this.previous = previous;
  }

  … gets e sets
}
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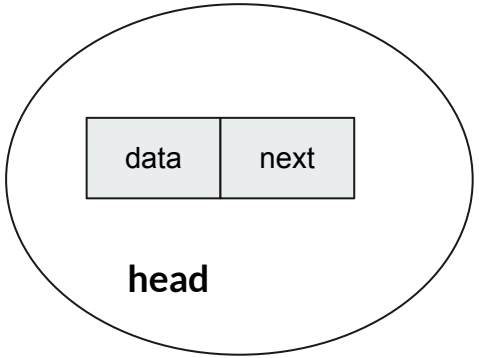
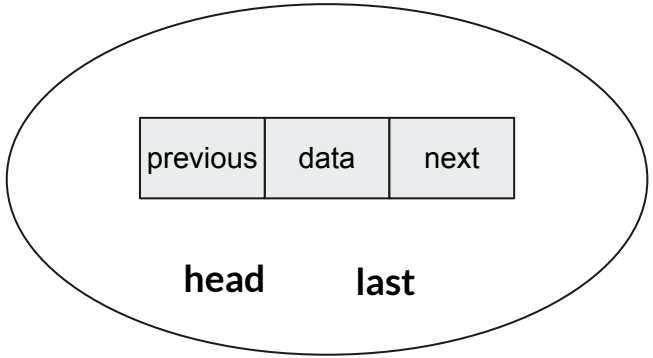
Listas ligadas - implementação



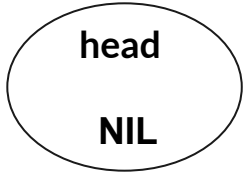


Listas ligadas (simples) - isEmpty

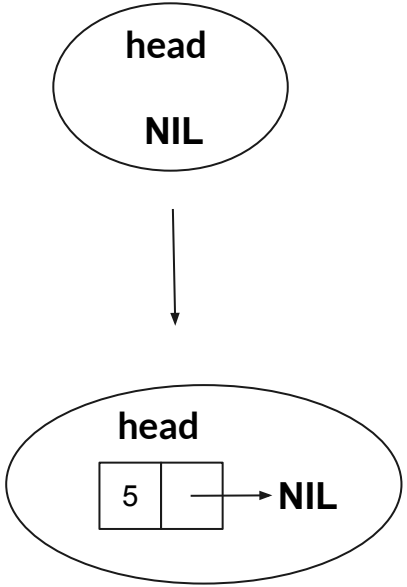
```
isEmpty( ){
  return head == NIL
}
```

Listas ligadas (simples) - insert

insert(5)



Listas ligadas (simples) - insert



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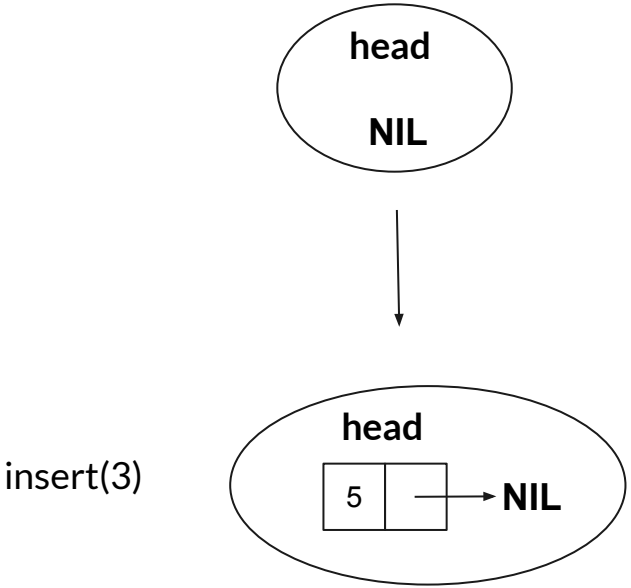
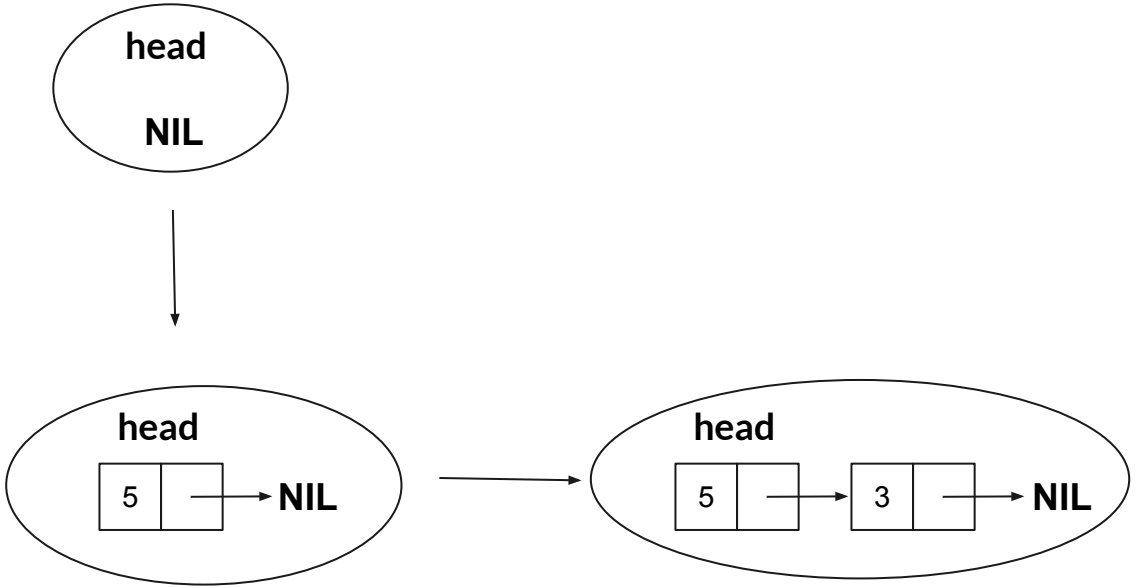


Listas ligadas (simples) - insert



Listas ligadas (simples) - insert

```
list-insert(item){
  auxHead = head
  if(head = NIL){
    newHead = new SingleLinkedListNode(item)
    newHead.next= head
    head = newHead
  } else{
    while(auxHead.next != NIL){
       auxHead = auxHead.next
    }
    auxHead.data = item
    auxHead.next = new SingleLinkedListNode()
 }

}
```

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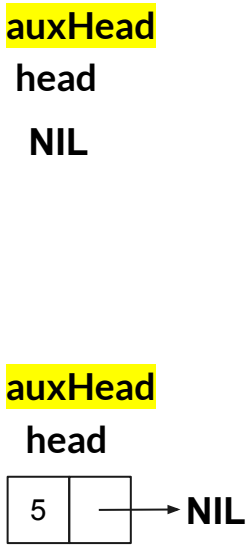
}
```



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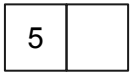
 }

}
```

auxHead

head

NIL



newHead

Listas ligadas (simples) - insert

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}
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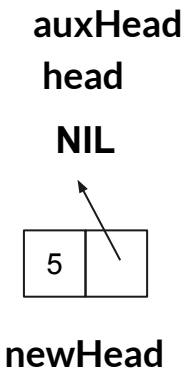

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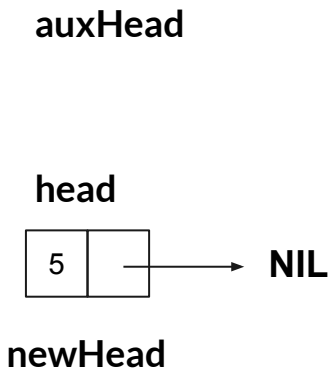

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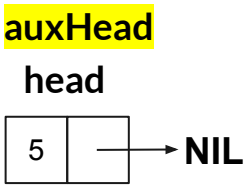

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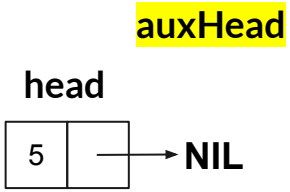

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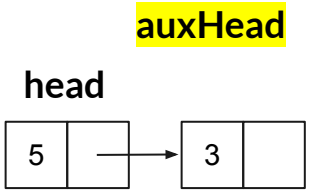

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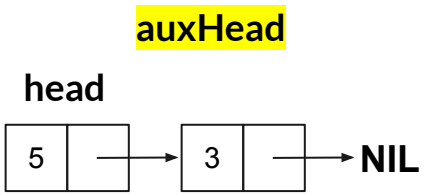

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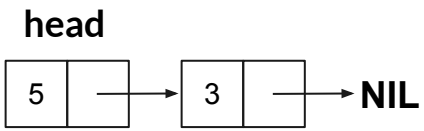
Qual a complexidade?



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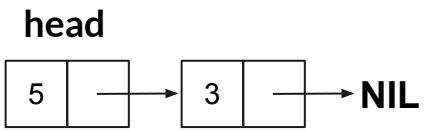
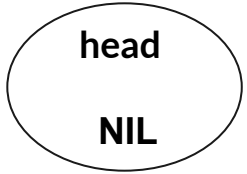
O(n)



Listas ligadas (simples) - search

search(3)



Listas ligadas (simples) - search

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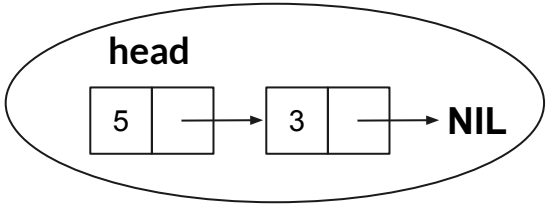

Listas ligadas (simples) - search

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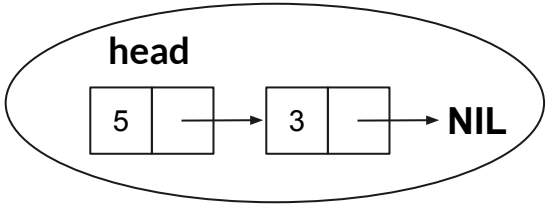
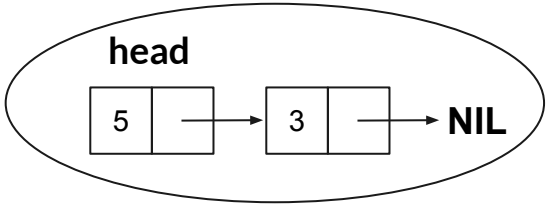



```
list-search(item){
  auxHead = head
  while(auxHead != NIL && auxHead.data != item){
      auxHead = auxHead.next
  }
  return auxHead
}
```

Listas ligadas (simples) - search

search(3)



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Listas ligadas (simples) - search

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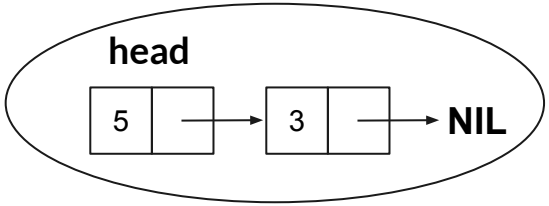

O(n)

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list-search(item){
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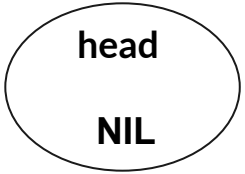
      auxHead = auxHead.next
  }
  return auxHead
}
```

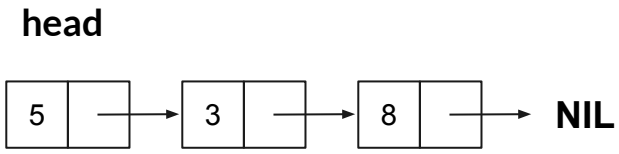
Listas ligadas (simples) - remove

remove(3)



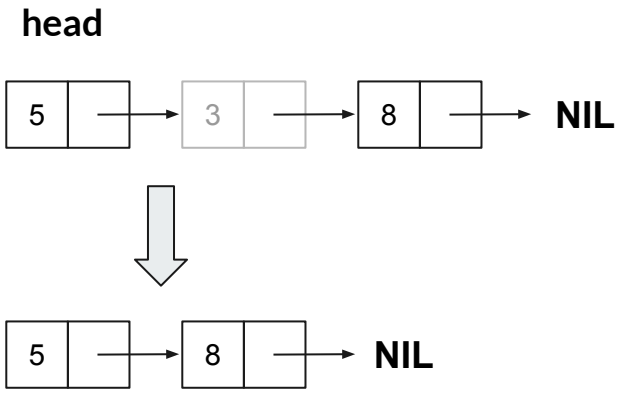
Listas ligadas (simples) - remove

remove(3)



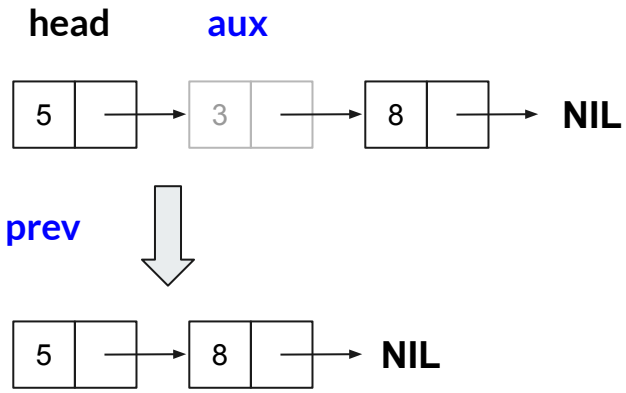
Listas ligadas (simples) - remove

remove(3)



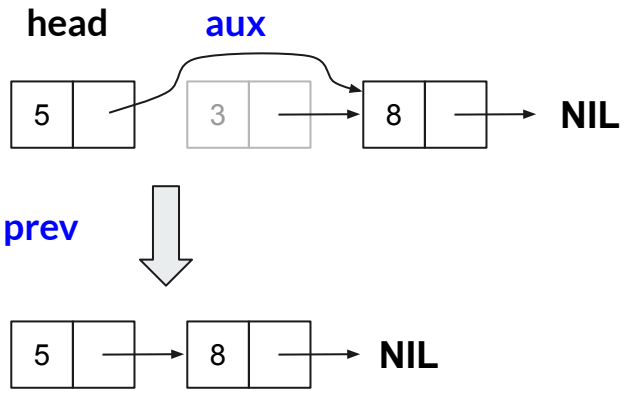
Listas ligadas (simples) - remove

remove(3)



Listas ligadas (simples) - remove

remove(3)



Listas ligadas (simples) - remove

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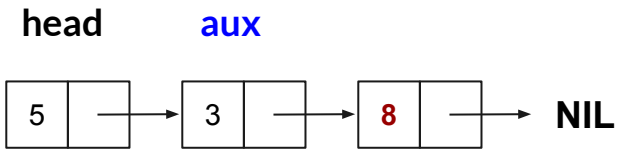

Listas ligadas (simples) - remove

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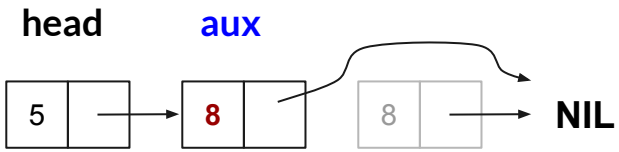


Listas ligadas (simples) - remove

```
list-remove(item){
  if head.data == item
     head = head.next
  else{
    aux = head
    while aux != NIL and aux.data != item {
      aux = aux.next
    }
    if aux != NIL{
     aux.data = aux.next.data
     aux.next = aux.next.next
  }
```

Listas ligadas (simples) - remove

```
list-remove(item){
  if head.data == item
     head = head.next
  else{
    aux = head
    while aux != NIL and aux.data != item {
      aux = aux.next
    }
    if aux != NIL{
     aux.data = aux.next.data
     aux.next = aux.next.next
  }
```

Qual a complexidade?

Listas ligadas (simples) - remove

```
list-remove(item){
  if head.data == item
     head = head.next
  else{
    aux = head
    while aux != NIL and aux.data != item {
      aux = aux.next
    }
    if aux != NIL{
     aux.data = aux.next.data
     aux.next = aux.next.next
  }
```

O(n)

Listas ligadas (simples) - size

```
list-size(){
  size = 0
  aux = head
  while aux != NIL {
      size = size + 1
      aux = aux.next
  }
  return size
}
```

Listas ligadas (simples) - size

```
list-size(){
  size = 0
  aux = head
  while aux != NIL {
      size = size + 1
      aux = aux.next
  }
  return size
}
```

Qual a complexidade?

Listas ligadas (simples) - size

```
list-size(){
  size = 0
  aux = head
  while aux != NIL {
      size = size + 1
      aux = aux.next
  }
  return size
}
```

O(n)

Listas duplamente ligadas

SingleLinkedListImpl

DoubleLinkedListImpl

```
public class DoubleLinkedListImpl<T>
    extends SingleLinkedListImpl<T>
     implements DoubleLinkedList<T>{

  protected DoubleLinkedListNode<T> last;

  public void insertFirst(T element)
  public void removeFirst()
  public void removeLast()

  ...
}
```

Listas duplamente ligadas

SingleLinkedListImpl

DoubleLinkedListImpl

Algo herdado precisa ser sobrescrito?

```
public class DoubleLinkedListImpl<T>
    extends SingleLinkedListImpl<T>
     implements DoubleLinkedList<T>{

  protected DoubleLinkedListNode<T> last;

  public void insertFirst(T element)
  public void removeFirst()
  public void removeLast()

  ...
}
```

Listas duplamente ligadas

SingleLinkedListImpl

DoubleLinkedListImpl

search (otimização)
insert
remove

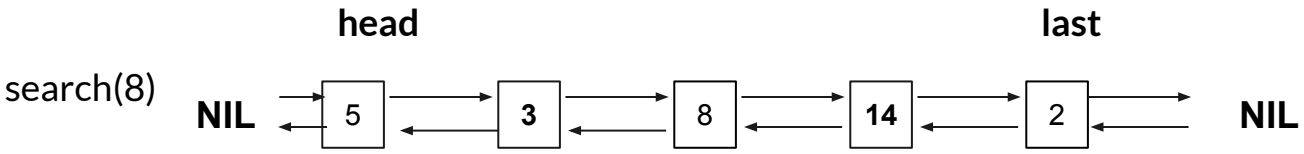
```
public class DoubleLinkedListImpl<T>
    extends SingleLinkedListImpl<T>
     implements DoubleLinkedList<T>{

  protected DoubleLinkedListNode<T> last;

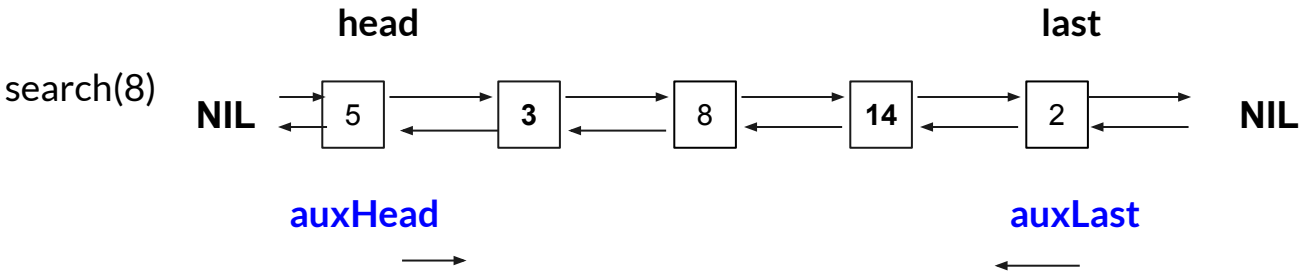
  public void insertFirst(T element)
  public void removeFirst()
  public void removeLast()

  ...
}
```

Listas duplamente ligadas - search



Listas duplamente ligadas - search



Listas duplamente ligadas - search

```
List-search(item){
  auxHead = head
  auxLast = last
  while(auxHead != auxLast and auxHead.next != auxLast
          and auxHead.data != item and auxLast.data != item){
    auxHead = auxHead.next
    auxLast = auxLast.previous
  }
  if auxHead.data == item
     return auxHead.data
  if auxLast.data == item
     return auxLast.data
}
```

Listas duplamente ligadas - search

```
List-search(item){
  auxHead = head
  auxLast = last
  while(auxHead != auxLast and auxHead.next != auxLast
          and auxHead.data != item and auxLast.data != item){
    auxHead = auxHead.next
    auxLast = auxLast.previous
  }
  if auxHead.data == item
     return auxHead.data
  if auxLast.data == item
     return auxLast.data
}
```

Qual a complexidade?

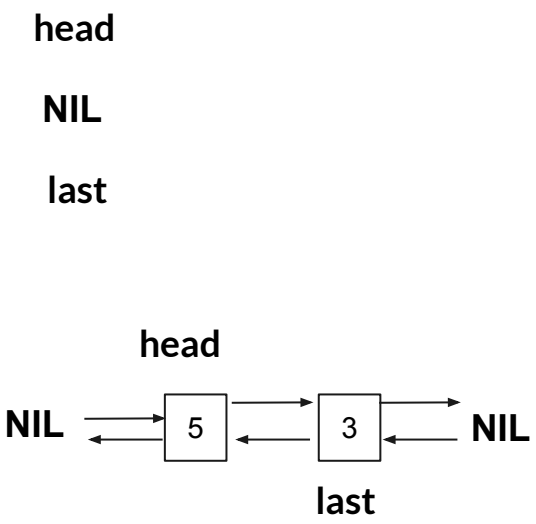
Listas duplamente ligadas - search

```
List-search(item){
  auxHead = head
  auxLast = last
  while(auxHead != auxLast and auxHead.next != auxLast
          and auxHead.data != item and auxLast.data != item){
    auxHead = auxHead.next
    auxLast = auxLast.previous
  }
  if auxHead.data == item
     return auxHead.data
  if auxLast.data == item
     return auxLast.data
}
```

O(n)

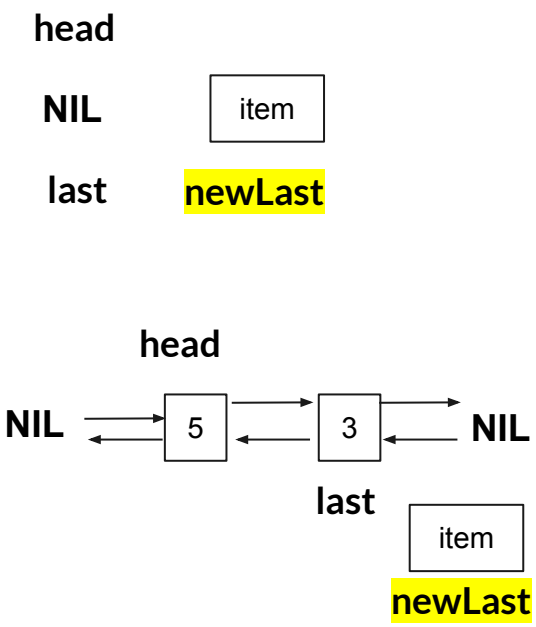
Listas duplamente ligadas - insert (final)

```
insertEnd(item){
  newLast = new DoubleLinkedListNode(item)
  newLast.previous = last
  newLast.next = NIL
  last.next = newLast
  if last == NIL{ //para o caso da lista vazia
    head = newLast
  }
  last = newLast
}
```



Listas duplamente ligadas - insert (final)

```
insertEnd(item){
  newLast = new DoubleLinkedListNode(item)
  newLast.previous = last
  newLast.next = NIL
  last.next = newLast
  if last == NIL{ //para o caso da lista vazia
    head = newLast
  }
  last = newLast
}
```



Listas duplamente ligadas - insert (final)

```
insertEnd(item){
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  newLast.previous = last
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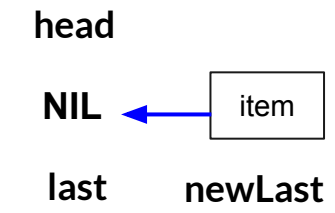
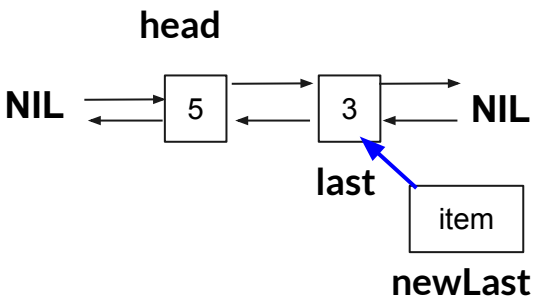
  last.next = newLast
  if last == NIL{ //para o caso da lista vazia
    head = newLast
  }
  last = newLast
}
```





Listas duplamente ligadas - insert (final)

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insertEnd(item){
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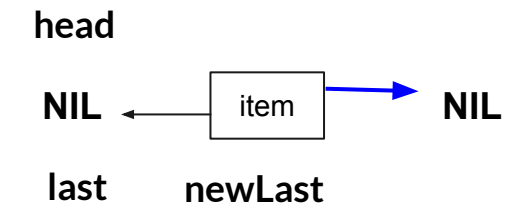
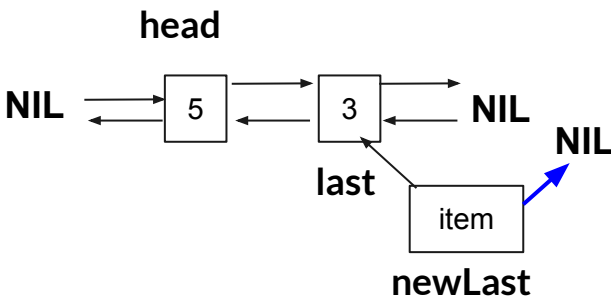
  }
  last = newLast
}
```





Listas duplamente ligadas - insert (final)

```
insertEnd(item){
  newLast = new DoubleLinkedListNode(item)
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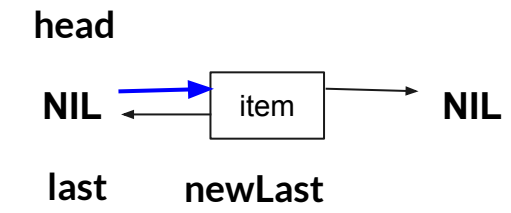
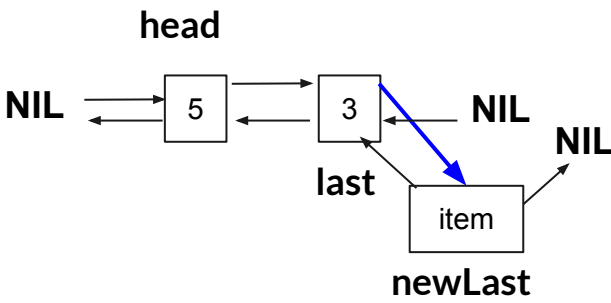
  if last == NIL{ //para o caso da lista vazia
    head = newLast
  }
  last = newLast
}
```





Listas duplamente ligadas - insert (final)

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}
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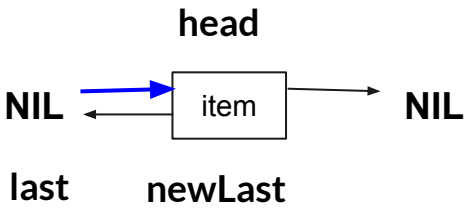

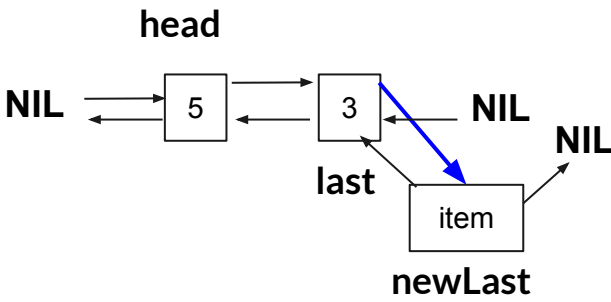

Listas duplamente ligadas - insert (final)

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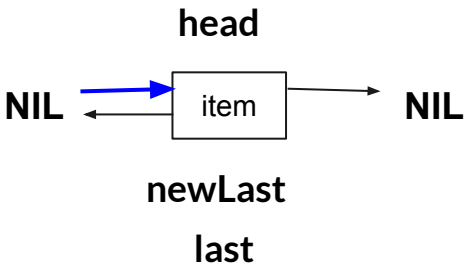
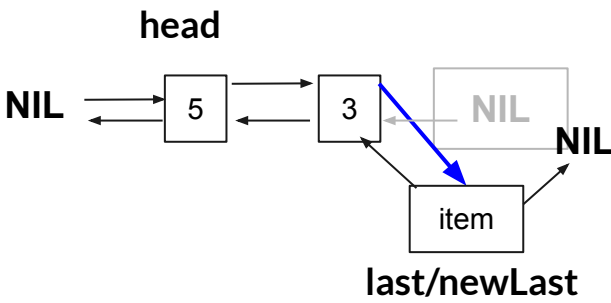
  }
  last = newLast
}
```





Listas duplamente ligadas - insert (final)

```
insertEnd(item){
  newLast = new DoubleLinkedListNode(item)
  newLast.previous = last
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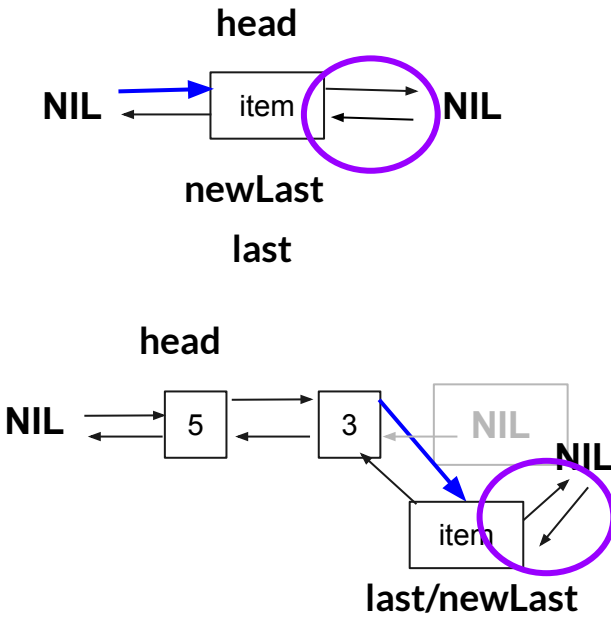
  if last == NIL{ //para o caso da lista vazia
    head = newLast
  }
  last = newLast
}
```



Listas duplamente ligadas - insert (final)

```
insertEnd(item){
  newLast = new DoubleLinkedListNode(item)
  newLast.previous = last
  newLast.next = NIL
  last.next = newLast
  if last == NIL{ //para o caso da lista vazia
    head = newLast
  }
  last = newLast
}
```

O(1)

Listas duplamente ligadas - insertFirst

```
insertFirst(item){
  newHead = new DoubleLinkedListNode(item)
  newHead.next = head
  newHead.previous = NIL
  head.previous = newHead
  if head == NIL{  //para o caso da lista vazia
    last = newHead
  }
 head = newHead
}
```

O(1)

Espelho do insertEnd, mas na outra extremidade

Listas duplamente ligadas - remove (pela chave)

```
list-remove(item){
  if head.data == item
     remove-first()
  else{
    aux = head
    while aux != NIL and aux.data != item {
      aux = aux.next
    }
    if aux != NIL{
     aux.previous.next = aux.next
     aux.next.previous = aux.previous
  }
```

Mesma lógica do remove da lista simples, exceto que a manipulação dos apontadores é simplificada

Listas duplamente ligadas - remove (pela chave)

```
list-remove(item){
  if head.data == item
     remove-first()
  else{
    aux = head
    while aux != NIL and aux.data != item {
      aux = aux.next
    }
    if aux != NIL{
     aux.previous.next = aux.next
     aux.next.previous = aux.previous
  }
```

Listas duplamente ligadas - remove (pela chave)

```
list-remove(item){
  if head.data == item
     remove-first()
  else{
    aux = head
    while aux != NIL and aux.data != item {
      aux = aux.next
    }
    if aux != NIL{
     aux.previous.next = aux.next
     aux.next.previous = aux.previous
  }
```

Posiciona aux no nó a ser removido ou NIL (final da lista)

Listas duplamente ligadas - remove (pela chave)

```
list-remove(item){
  if head.data == item
     remove-first()
  else{
    aux = head
    while aux != NIL and aux.data != item {
      aux = aux.next
    }
    if aux != NIL{
     aux.previous.next = aux.next
     aux.next.previous = aux.previous
  }
```

Se aux aponta para nó com dado então usa previous e next para ajustar a lista convenientemente

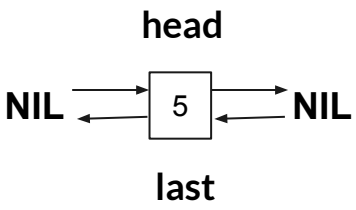
Listas duplamente ligadas - remove (pela chave)

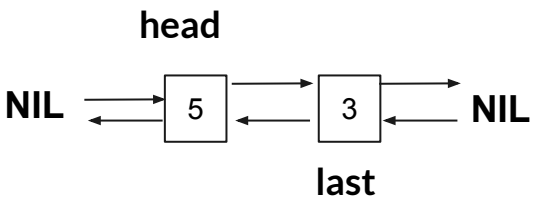
```
list-remove(item){
  if head.data == item
     remove-first()
  else{
    aux = head
    while aux != NIL and aux.data != item {
      aux = aux.next
    }
    if aux != NIL{
     aux.previous.next = aux.next
     aux.next.previous = aux.previous
  }
```

O(n)

Listas duplamente ligadas - removeFirst

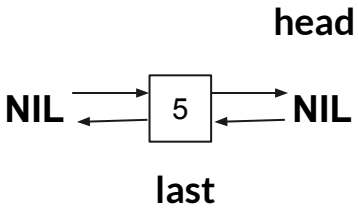
```
removeFirst(item){
  if head != NIL {
    head = head.next
    if head == NIL
      last = head
    else
      head.previous = NIL
  }
}
```

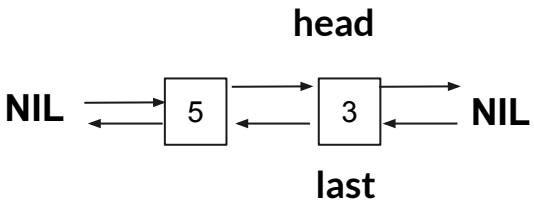




Listas duplamente ligadas - removeFirst

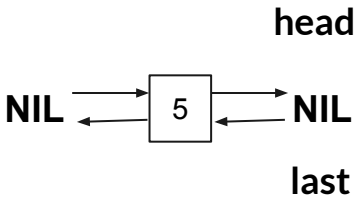
```
removeFirst(item){
  if head != NIL {
    head = head.next
    if head == NIL
      last = head
    else
      head.previous = NIL
  }
}
```

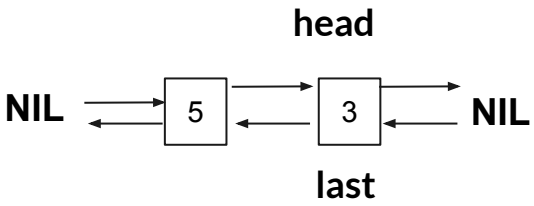




Listas duplamente ligadas - removeFirst

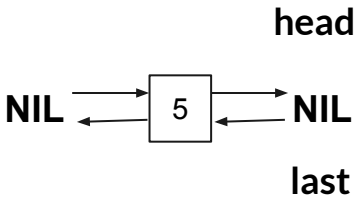
```
removeFirst(item){
  if head != NIL {
    head = head.next
    if head == NIL
      last = head
    else
      head.previous = NIL
  }
}
```

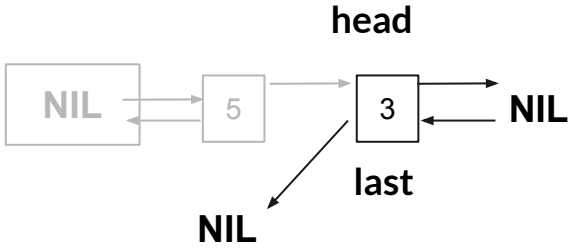




Listas duplamente ligadas - removeFirst

```
removeFirst(item){
  if head != NIL {
    head = head.next
    if head == NIL
      last = head
    else
      head.previous = NIL
  }
}
```





Listas duplamente ligadas - removeFirst

```
removeFirst(item){
  if head != NIL {
    head = head.next
    if head == NIL
      last = head
    else
      head.previous = NIL
  }
}
```

O(1)

Listas duplamente ligadas - removeEnd

```
removeEnd(item){
  if last != NIL {
    last = last.previous
    if last == NIL
      head = last
    else
      last.next = NIL
  }
}
```

Espelho do removeFirst, mas na outra extremidade

O(1)

Listas encadeadas (Java)

Class LinkedList<E>

```
java.lang.Object
    java.util.AbstractCollection<E>
        java.util.AbstractList<E>
            java.util.AbstractSequentialList<E>
                java.util.LinkedList<E>
```

Type Parameters:

`E - the type of elements held in this collection`

All Implemented Interfaces:

`Serializable, Cloneable, Iterable<E>, Collection<E>, Deque<E>, List<E>, Queue<E>`

Listas encadeadas

Aplicações e exemplos:

- Implementar uma lista ordenada (possivelmente com Comparator)
- Implementar uma pilha usando listas
- Implementar uma fila usando listas
- Implementar ordenação em uma lista ligada

Lista Encadeada
Abordagem Recursiva

Lista Encadeada

Considerar um nó como sendo a própria lista

Implementação recursiva dos métodos (maior parte)

Interface continua a mesma!

Listas ligadas - implementação (antes)

```
public interface LinkedList<T> {
  public void insert(T element);
  public void remove(T element);
  public T search(T element);
  public int size();
  public boolean isEmpty();
  public T[] toArray();
}
public interface DoubleLinkedList<T>
      extends LinkedList<T> {

  public void insertFirst(T element);
  public void removeFirst();
  public void removeLast();

}
```

```
public class SingleLinkedListImpl<T>
      implements LinkedList<T> {
  protected SingleLinkedListNode<T> head;
  ...
}
public class DoubleLinkedListImpl<T> extends
   SingleLinkedListImpl<T> implements DoubleLinkedList<T> {
  protected DoubleLinkedListNode<T> last;
  ...
}
public class SingleLinkedListNode<T> {
  protected T data;
  protected SingleLinkedListNode<T> next;
}
public class DoubleLinkedListNode<T>
      extends SingleLinkedListNode<T> {
  protected DoubleLinkedListNode<T> previous;
}
```

Listas ligadas - implementação (depois)

```
public interface LinkedList<T> {
  public void insert(T element);
  public void remove(T element);
  public T search(T element);
  public int size();
  public boolean isEmpty();
  public T[] toArray();
}
public interface DoubleLinkedList<T>
      extends LinkedList<T> {

  public void insertFirst(T element);
  public void removeFirst();
  public void removeLast();

}
```

```
public class RecursiveSingleLinkedListImpl<T>
      implements LinkedList<T> {

  protected T data;
  protected RecursiveSingleLinkedListImpl<T> next;

  public RecursiveSingleLinkedListImpl() { }
  ...
}
public class RecursiveDoubleLinkedListImpl<T> extends
    RecursiveSingleLinkedListImpl<T> implements
    DoubleLinkedList<T> {

  protected RecursiveDoubleLinkedListImpl<T> previous;

  public RecursiveDoubleLinkedListImpl() {  }
  ...
}
```

Listas ligadas - implementação (depois)

```
public interface LinkedList<T> {
  public void insert(T element);
  public void remove(T element);
  public T search(T element);
  public int size();
  public boolean isEmpty();
  public T[] to
}
public interfac
     extends Li

  public void i
  public void r
  public void removeLast();

}
```

```
public class RecursiveSingleLinkedListImpl<T>
      implements LinkedList<T> {

  protected T data;
  protected RecursiveSingleLinkedListImpl<T> next;

  public RecursiveSingleLinkedListImpl() { }

                                  ol<T> extends
                                  lements

                                  l<T> previous;

  public RecursiveDoubleLinkedListImpl() {  }
  ...
}
```

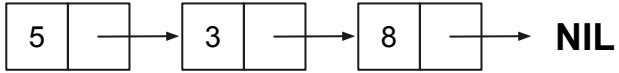
Métodos recursivos seguir o template:

- **Caso base (lista vazia/situação trivial)**
- **Caso indutivo (lista não vazia/situação não-trivial)**

Listas ligadas - implementação

NIL

caso base



caso indutivo

```
isEmpty(){
  if data == null //caso base
      return true
  else //caso indutivo
     return false
}
```

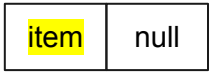
Listas ligadas - implementação

null	null

caso base

```
list-insert(item){
   if(isEmpty) //caso base
      data = item
      next = new RecursiveSingleLinkedListImpl<T>( )
   else //caso indutivo
      next.list-insert(item)
}
```

Listas ligadas - implementação



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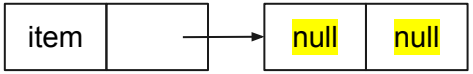
Listas ligadas - implementação



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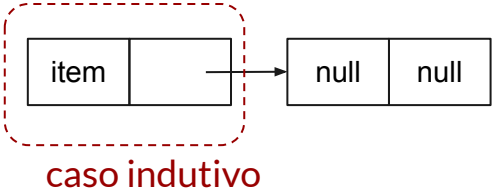
Listas ligadas - implementação

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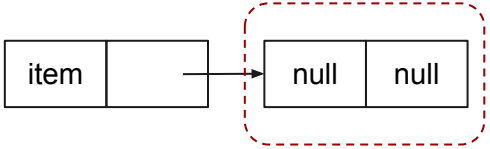
Listas ligadas - implementação



caso indutivo

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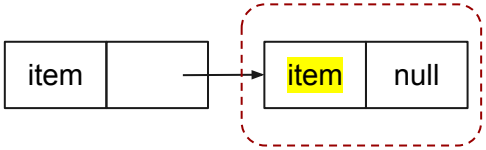
Listas ligadas - implementação



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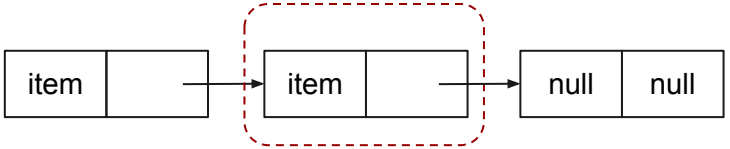
Listas ligadas - implementação



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}
```

Listas ligadas - implementação

Qual a complexidade?

```
list-insert(item){
   if(isEmpty) //caso base
      data = item
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   else //caso indutivo
      next.list-insert(item)
}
```

Listas ligadas - implementação

Qual a complexidade?

```
list-insert(item){
   if(isEmpty) //caso base
      data = item
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   else //caso indutivo
      next.list-insert(item)
}
```

T(n)

O(1)

T(n - 1)

Listas ligadas - implementação

Qual a complexidade?

$$T(n) = T(n - 1) + O(1) \Rightarrow \theta(n)$$

```
list-insert(item){
   if(isEmpty) //caso base
      data = item
      next = new RecursiveSingleLinkedListImpl<T>( )
   else //caso indutivo
      next.list-insert(item)
}
```

T(n)

O(1)

T(n - 1)

Listas ligadas - implementação

```
list-search(item){
   if(isEmpty) //caso base

   else




}
```

Listas ligadas - implementação

```
list-search(item){
  if(isEmpty) //caso base
    return null
  else
    if data==item //caso base
      return data
    else //caso indutivo
      return next.list-search(item)
}
```

Listas ligadas - implementação

```
list-search(item){
  if(isEmpty) //caso base
    return null
  else
    if data==item //caso base
      return data
    else //caso indutivo
      return next.list-search(item)
}
```

$T(n) = T(n - 1) + O(1) \Rightarrow \theta(n)$

Listas ligadas - implementação

```
list-remove(item){
   if(isEmpty) //caso base
      //nao faz nada porque nao contem o elemento
   else //caso indutivo




}
```

Listas ligadas - implementação

```
list-remove(item){
   if(isEmpty) //caso base
      //nao faz nada porque nao contem o elemento
   else
         if data==item //caso base
         data = next.data
         next = next.next
      else //caso indutivo
         next.list-remove(item)
}
```

Listas ligadas - implementação

```
list-remove(item){
   if(isEmpty) //caso base
      //nao faz nada porque nao contem o elemento
   else
         if data==item //caso base
         data = next.data
         next = next.next
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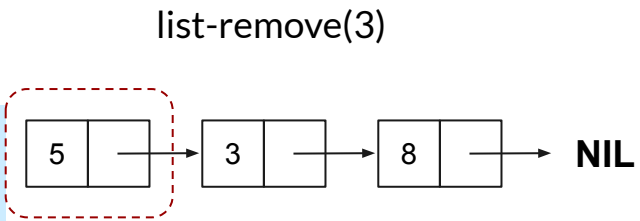
         next.list-remove(item)
}
```



Listas ligadas - implementação

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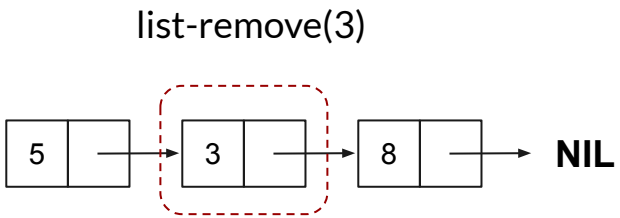

Listas ligadas - implementação

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list-remove(item){
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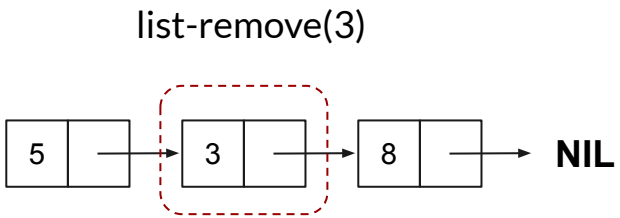
      //nao faz nada porque nao contem o elemento
   else
         if data==item //caso base
         data = next.data
         next = next.next
      else //caso indutivo
         next.list-remove(item)
}
```



Listas ligadas - implementação

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      //nao faz nada porque nao contem o elemento
   else
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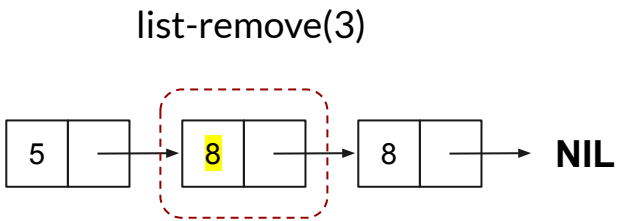
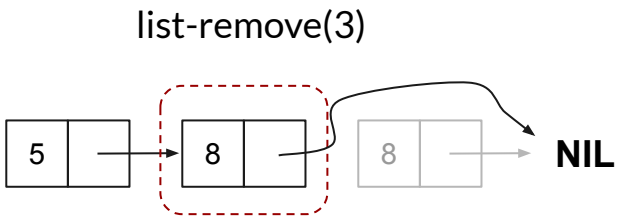
         data = next.data
         next = next.next
      else //caso indutivo
         next.list-remove(item)
}
```



Listas ligadas - implementação

```
list-remove(item){
   if(isEmpty) //caso base
      //nao faz nada porque nao contem o elemento
   else
         if data==item //caso base
         data = next.data
         next = next.next
      else //caso indutivo
         next.list-remove(item)
}
```



Listas ligadas - implementação

```
list-remove(item){
   if(isEmpty) //caso base
      //nao faz nada porque nao contem o elemento
   else
         if data==item //caso base
         data = next.data
         next = next.next
      else //caso indutivo
         next.list-remove(item)
}
```

$T(n) = T(n - 1) + O(1) \Rightarrow \theta(n)$

Listas ligadas - implementação

```
list-size(){
   if(isEmpty) //caso base

   else //caso indutivo

}
```

Listas ligadas - implementação

```
list-size(){
   if(isEmpty) //caso base
     return 0
   else //caso indutivo
     return 1 + next.list-size()
}
```

$T(n) = T(n - 1) + O(1) \Rightarrow \theta(n)$

Listas ligadas - implementação

```
T[] list-toArray(){
   T[] result = new java.util.LinkedList()
   toArray(result,this)
   return (T[]) result.toArray()
}

toArray (LinkedList list, RecursiveSingleLinkedListImpl node){
   if (!node.isEmpty){
      list.add(node.data)
      toArray(list,node.next)
   }
}
```

Listas ligadas - implementação

Implemente um método que encontra o maior elemento de uma lista ligada

Listas ligadas - implementação

Implemente um método que encontra o maior elemento de uma lista ligada

```
maior(){
  if(isEmpty) //caso base
    return 0
  else //caso indutivo
    maior = this.data
    maiorNext = next.maior()
    if(maiorNext > maior)
        maior = maiorNext
    return maior
}
```

Listas duplamente ligadas

RecursiveSingleLinkedListImpl

RecursiveDoubleLinkedListImpl

insert
remove

```
public class RecursiveSingleLinkedListImpl<T>
      implements LinkedList<T> {

  protected T data;
  protected RecursiveSingleLinkedListImpl<T> next;

  public RecursiveSingleLinkedListImpl() { }
  ...
}
public class RecursiveDoubleLinkedListImpl<T> extends
    RecursiveSingleLinkedListImpl<T> implements
    DoubleLinkedList<T> {

  protected RecursiveDoubleLinkedListImpl<T> previous;

  public RecursiveDoubleLinkedListImpl() {  }
  ...
}
```

Listas duplamente ligadas - insert (herdado)

```
list-insert(item){
   if(isEmpty) //caso base
      data = item
      next = NIL
   else //caso indutivo
      next.list-insert(item)
}
```

Listas duplamente ligadas - insert (sobrescrito)

```
list-insert(item){
   if(isEmpty) //caso base
      data = item
      next = NIL
     if (previous == null)
         previous = NIL
   else //caso indutivo
      next.list-insert(item)
}
```

Listas duplamente ligadas - remove (herdado)

```
list-remove(item){
  if(isEmpty) //caso base
    //nao faz nada porque nao contem o elemento
  else
    if data==item //caso base
      data = next.data
      next = next.next
    else //caso indutivo
      next.list-remove(item)
}
```

Listas duplamente ligadas - remove (sobrescrito)

```
list-remove(item){
  if(isEmpty) //caso base
    //nao faz nada porque nao contem o elemento
  else
    if data==item //caso base
      if(previous.isEmpty && next.isEmpty){
        data = next = previous = null
      else
        data = next.data
        next = next.next
        if(next != null)
          next.previous = this
    else //caso indutivo
      next.list-remove(item)
}
```

Listas duplamente ligadas

```
public void insertFirst(T element)
public void removeFirst();
public void removeLast()
```

Quais precisam ser recursivos?

Listas duplamente ligadas

```
public void insertFirst(T element)
public void removeFirst();
public void removeLast()
```

Quais precisam ser recursivos?

Implementações deixadas como exercício